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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS 1-1 **Date:** 09/16/2026

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If

you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, Developing, Testing, and Maintaining software applications and systems.	Software Developer, Full-Stack Engineer
Systems & Cloud Infrastructure	Managing, Deploying, and Maintaining servers, networks, cloud architectures, and system operations.	Cloud Engineer, Systems Administrator
Data Science & AI Engineering	Extracting insights from data, building predictive models, and developing artificial intelligence applications.	Data Analyst, Machine Learning Engineer
Cybersecurity & Threat Defense	Protecting digital assets, networks, and applications from cyber threats and unauthorized access.	Cybersecurity Analyst, Penetration Tester
Information Systems & Enterprise IT	Aligning technology solutions with business goals, managing IT infrastructure, and overseeing enterprise software.	IT Business Analyst, Systems Manager.

My Chosen Pathway (Part B)

I am most interested in: Software Engineering

Why I chose it:

I've always been interested in how apps and tools are built from scratch, and I really enjoy the process of solving step-by-step logic problems. The idea of writing code that creates useful software for people to use every day is what excites me most about tech.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Junior Software Developer, Associate Frontend Developer	Writing clean code, fixing bugs, testing components, version control, and learning codebase.
Mid-Level	Software Engineer, Full-Stack Developer	Designing features, API integration, code reviews, database management, and optimizing performance.
Senior-Level	Senior Software Engineer, Tech Lead	System architecture design, mentoring juniors, leading development teams, and technical decision-making.
Expert / Leadership	Software Architect, Chief Technology Officer (CTO)	High-level enterprise architecture, technology strategy, cross-team alignment, and driving innovation.

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Programming Fundamentals (Python)	N	Keep practicing with CC 101 lab tasks and practice small coding challenges on my own.
Data Structures and Algorithms	Y	Take our upcoming programming discussions seriously and supplement with free online courses or tutorials
Version Control (GitHub)	Y	Currently using GitHub basics but still learning and relying on tutorials.
Database Management	N	Plan to practice through free tutorials before taking our database courses.
Web Development Frameworks	N	Build simple personal web projects during free time to get comfortable with basic frameworks.

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Junior Web Developer
Company	Accenture Philippines
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	JobStreet
Required Skills Listed (3–5)	HTML/CSS/JavaScript,RESTful API basics, Git version control, basic SQL database operations

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I chose Software Engineering because it directly connects to why I wanted to take BSCS in the first place. I want to build functional applications.

2. Looking at your roadmap, which career stage excites you the most, and why?

The Mid-Level Software Engineer stage excites me the most because that's where you get to work independently on full features and solve bigger technical problems. It feels like a sweet spot where you have enough foundation to build complete systems without being bogged down purely by high-level management duties.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

I already have basic familiarity with programming logic from my basic studies. However, mastering Data Structures and Algorithms will likely be the hardest challenge for me because it requires abstract logic and mathematical thinking that takes a lot of practice to get right.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

Mapping this roadmap confirms that sticking strictly to classroom lectures isn't enough. I need to build my own personal project that connects to my coursework. It gives me a goal to focus on each semester so I can gradually build the practical portfolio required for an entry-level position when I graduate.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is a bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date